EC3SD03
Tutorial 1

Topics

- Taylor Series
- Solution of Nonlinear Equation

Taylor Series - Maclaurin Series

$$f(x) \cong f(x_i) + \frac{f'(x_i)}{1!}(x - x_i) + \frac{f''(x_i)}{2!}(x - x_i)^2 + \ldots + \frac{f^{(n)}(x_i)}{n!}(x - x_i)^n + R_n$$

M

$$f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$

Example: Problem 4.1

4.1 The Maclaurin series expansion for cos x is

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$$

Starting with the simplest version, cos $x = 1$, add terms one at a time to estimate $\cos(\pi/3)$. After each new term is added, compute the true and approximate percent relative errors. Use your pocket calculator to determine the true value. Add terms until the absolute value of the approximate error estimate falls below an error criterion conforming to two significant figures.

Solution

Estimate $\cos(\pi/3)$ using the Maclaurin series expansion:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$x = \frac{\pi}{3} = 1.04719755$$

True value (calculator):

$$\cos(\pi/3) = 0.5$$

if the following criterion is met, we can be assured that the result is correct to *at least n* significant figures.

$$\varepsilon_s = (0.5 \times 10^{2-n})\ \%$$

Stopping Criterion (Two Significant Figures)

For two significant figures:

$$\varepsilon_s = 0.5 \times 10^{2-2} = 0.5\%$$

We stop when

$$|\varepsilon_a| < 0.5\%$$

Step 1: Successive Approximations

0th Approximation

$$S_0 = 1$$

$$\varepsilon_t = -100\%$$

1st term

$$S_1 = 1 - \frac{x^2}{2}$$

$$x^2 = 1.09662$$

$$S_1 = 1 - 0.54831 = 0.45169$$

True % error:

$$\varepsilon_t = \frac{0.5 - 0.45169}{0.5} \times 100 = 9.662\%$$

Approx. % error:

$$\varepsilon_a = \frac{0.45169 - 1}{0.45169} \times 100 = -121.37\%$$

Approximation	S_n Value	True % Error	Approx. % Error
S0	1.00000	100%	---
S1	0.45169	9.662%	121.37%
S2	0.50180	0.36%	9.98%
S3	0.49997	0.006%	0.366%

$$|\varepsilon_a| = 0.366\% < 0.5\%$$

The stopping criterion is satisfied.

Problem 2 (from Previous Midterm Exam)

Given

$$f(x) = \sqrt{x}$$

Use the Taylor series about $a = 2$ to approximate $f(2.5)$ with a relative true error less than 0.05%.

Step 1: Find the derivatives

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$$f(x) = x^{1/2}$$

$$f'(x) = \frac{1}{2}x^{-1/2}$$

$$f''(x) = -\frac{1}{4}x^{-3/2}$$

$$f'''(x) = \frac{3}{8}x^{-5/2}$$

$$f^{(4)}(x) = -\frac{15}{16}x^{-7/2}$$

Derivatives at $a = 2$

Quantity	Value
$f(2)$	1.4142
$f'(2) = \frac{1}{2\sqrt{2}}$	0.3536
$f''(2) = -\frac{1}{4(2)^{3/2}}$	-0.0884
$f'''(2) = \frac{3}{8(2)^{5/2}}$	0.0663
$f^{(4)}(2) = -\frac{15}{16(2)^{7/2}}$	-0.0829

Order	Calculation	Approximation
T_0	$f(2)$	1.4142
T_1	$1.4142 + (0.3536)(0.5)$	1.5910
T_2	$1.5910 + \frac{-0.0884}{2}(0.5)^2$	1.5800
T_3	$1.5800 + \frac{0.0663}{6}(0.5)^3$	1.5814
T_4	$1.5814 + \frac{-0.0829}{24}(0.5)^4$	1.5812

Using the true value

$$f(2.5) = \sqrt{2.5} = 1.5811$$

(rounded to 4 decimal places), the **relative true error** at each iteration is

$$\varepsilon_t = \left| \frac{\text{True Value} - \text{Approximation}}{\text{True Value}} \right| \times 100$$

Iteration	Relative True Error (%)	Meets $< 0.05\%$?
T_0	10.5566	No
T_1	0.6261	No
T_2	0.0696	No
T_3	0.0190	Yes
T_4	0.0063	Yes

Therefore, the **third-order Taylor polynomial** T_3 is the first approximation that satisfies the required accuracy:

$$\boxed{\sqrt{2.5} \approx 1.5814}$$

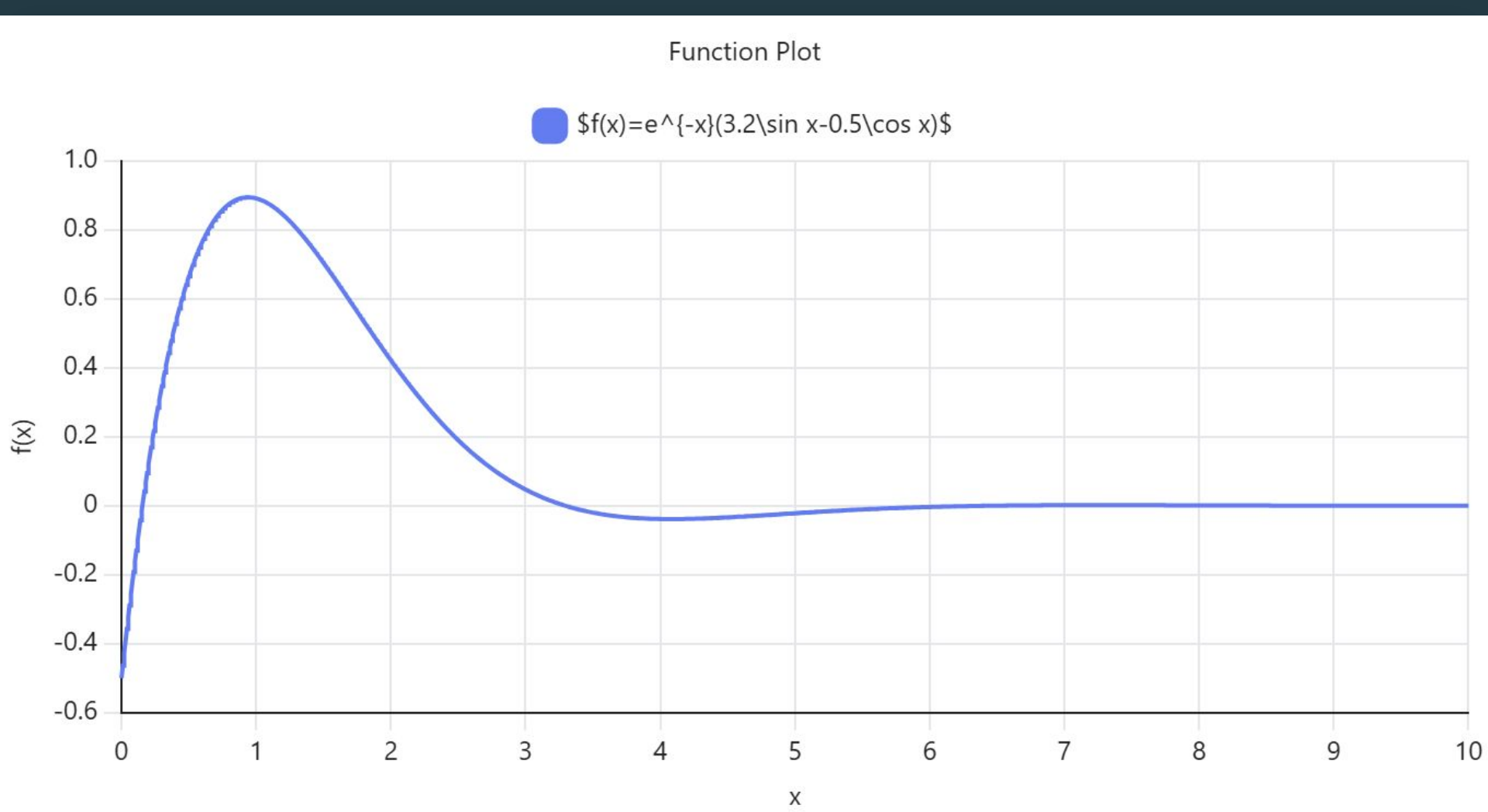
with

$$\boxed{\varepsilon_t = 0.0190\% < 0.05\%}.$$

Bracketing Method

Consider $$f(x) = e^{-x}(3.2\sin(x) - 0.5\cos(x))$$

Use the false position method to solve the equation on the interval [3, 4] (3 iterations). Calculate the relative approximate error at each iteration.


Function Plot
$f(x)=e^{-x}(3.2\sin x-0.5\cos x)$
1.0
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
f(x)
0
1
2
3
4
5
6
7
8
9
10
x

Using false position

$$x_m = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$

$$f(3) = 0.0465, \qquad f(4) = -0.0396$$

Since $f(3)f(4) < 0$, a root exists in $[3, 4]$.

#	x_L	f(x_L)	x_u	$f(x_u)$	x_m	$f(x_m)$	RAE
1	3	0.04713	4	−0.03837	3.551	−0.02339	-
2	3	0.04713	3.551	−0.02339	3.368	−0.007963	5.433 %
3	3	0.04713	3.368	−0.007963	3.315	−0.002166	1.599 %

For the previous function, apply bracketing methods (**bisection method, false position method, modified false position method)** for **5 iterations** and compare the results

1. Bisection Method (5 Iterations)

n	a_n	$f(a_n)$	b_n	$f(b_n)$	c_n	$f(c_n)$	ε_a
1	3.0000	+0.0471	4.0000	-0.0384	3.5000	-0.0198	—
2	3.0000	+0.0471	3.5000	-0.0198	3.2500	+0.0058	7.69%
3	3.2500	+0.0058	3.5000	-0.0198	3.3750	-0.0087	3.70%
4	3.2500	+0.0058	3.3750	-0.0087	3.3125	-0.0019	1.89%
5	3.2500	+0.0058	3.3125	-0.0019	3.2812	+0.0019	0.95%

2. False Position Method (5 Iterations)

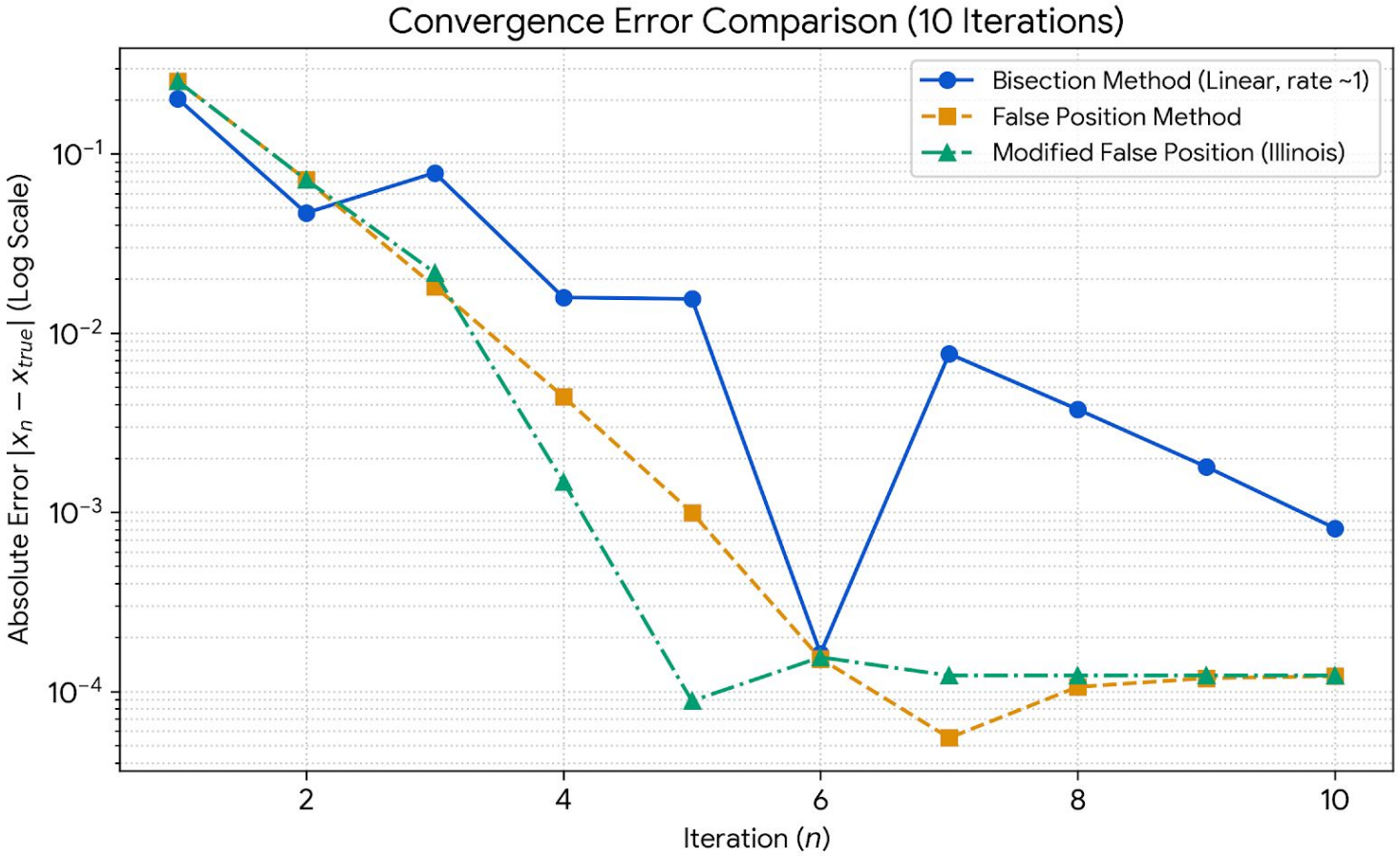
n	a□	f(a□)	b□	f(b□)	c□	f(c□)	ε_a
1	3.0000	+0.0471	4.0000	-0.0384	3.5512	-0.0234	—
2	3.0000	+0.0471	3.5512	-0.0234	3.3683	-0.0080	5.43%
3	3.0000	+0.0471	3.3683	-0.0080	3.3149	-0.0022	1.61%
4	3.0000	+0.0471	3.3149	-0.0022	3.3011	-0.0005	0.42%
5	3.0000	+0.0471	3.3011	-0.0005	3.2977	-0.0001	0.10%

3. Modified False Position / Illinois Method (5 Iterations)

n	a_n	$f(a_n)$	b_n	$f(b_n)$	c_n	$f(c_n)$	ε_a
1	3.0000	+0.0471	4.0000	-0.0384	3.5512	-0.0234	—
2	3.0000	+0.0471	3.5512	-0.0234	3.3683	-0.0080	5.43%
3	3.0000	+0.0236	3.3683	-0.0080	3.2750	+0.0026	2.85%
4	3.2750	+0.0026	3.3683	-0.0080	3.2982	-0.0002	0.70%
5	3.2750	+0.0026	3.2982	-0.0002	3.2966	-0.0000	0.05%

Final Iteration (n = 5) Summary

Method	a_5	$f(a_5)$	b_5	$f(b_5)$	c_5	$f(c_5)$	ε_a
Bisection	3.2500	+0.0058	3.3125	-0.0019	3.2812	+0.0019	0.95%
False Position	3.0000	+0.0471	3.3011	-0.0005	3.2977	-0.0001	0.10%
Modified False Position	3.2750	+0.0026	3.2982	-0.0002	3.2966	-0.0000	0.05%


Convergence Error Comparison (10 Iterations)
Bisection Method (Linear, rate ~1)
False Position Method
Modified False Position (Illinois)
Absolute Error $|x_n - x_{true}|$ (Log Scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
2
4
6
8
10
Iteration (n)

- **Bisection Method Performance:** Bisection (solid blue line) shows a steady, guaranteed reduction, though its error can oscillate depending on whether the midpoint happens to fall closer or further from the exact root in that specific iteration. Its reduction factor is strictly bounded by a factor of 2 ($\frac{1}{2}$) per iteration.
- **False Position vs. Modified False Position:**
  - Standard False Position (dashed orange line) achieves very rapid initial error reduction because secant lines capture the slope of the curve well on this interval. However, it can suffer from slow convergence or stagnation when one endpoint gets stuck due to curvature.
  - Modified False Position / Illinois Method (dash-dotted green line) drops sharply in the early iterations (reaching high precision around iterations 5–6) because halving the endpoint function value successfully breaks any one-sided stagnation and forces the interval to shrink much faster.

Summary

- In standard **False Position**, the left endpoint (a = 3.0) remained completely stuck for all 5 iterations because f(3.0) stayed large and positive, forcing all adjustments to happen on the upper bound.
- **Modified False Position (Illinois Algorithm)** successfully detected this stagnation at Iteration 3 by halving f(a). This forced the lower bound to shift up to 3.2750 in Iteration 4, breaking the one-sided trap and driving the error down to an exceptionally tight a = 0.05 %by Iteration 5.

(from previous midterm exam)

$$f(x)=x^3-9x+2$$

use,the Newton-Raphson method to solve the equation (3 iterations, $x_0 = 0$). Calculate the relative approximate error at each iteration.

$f(x) = x^3 - 9x + 2$, starting at $x0 = 0$

$f'(x) = 3x^2 - 9$

$x_{i+1} = x_i - f(x_i) / f'(x_i)$

$RAE = |x_{i+1} - x_i| / x_{i+1} \times 100$

i	xi	f(xi)	f'(xi)	xi+1	|xi+1 − xi|	RAE
1	0	2	−9	0.22222	0.22222	100 %
2	0.22222	0.010974	−8.852	0.22346	0.00124	0.555 %
3	0.22346	≈ 0.000001	−8.850	**0.22346**	≈ 0.0000002	**≈ 0.0001 %**

Result: root ≈ 0.2235 with very fast (quadratic) convergence: 100 % → 0.555 % → 0.0001 %. The posted table rounds x2 to 0.2234, giving 0.537 % and 0.045 %; the unrounded x2 = 0.22346 gives the values above.

Using fixed point iteration (x0=0)

$$x^3 - 9x + 2 = 0$$

$$9x = x^3 + 2$$

$$x = \frac{x^3 + 2}{9}$$

$$g(x) = \frac{x^3 + 2}{9}.$$

Check convergence:

$$g'(x) = \frac{x^2}{3}.$$

For $0 \leq x \leq 1$,

$$|g'(x)| \leq \frac{1}{3} < 1.$$

Therefore the iteration converges.

Iteration n	x_n	x_{n+1}	$\varepsilon_a(\%)$
0	0.0000000000	0.2222222222	100.0000
1	0.2222222222	0.2234413961	0.5456
2	0.2234413961	0.2234621648	0.0093
3	0.2234621648	0.2234625190	0.0002

Since

$$\varepsilon_a = 0.0002\% < 0.005\%,$$

the stopping criterion is satisfied at iteration 3.

Iterations (4 Significant Digits)

Iteration n	x_n	Calculation of x_{n+1}	$\varepsilon_a(\%)$
0	0.0000	$(0^3+2)/9 = 0.2222$	-
1	0.2222	$(0.2222^3+2)/9 = 0.2234$	0.5370
2	0.2234	$(0.2234^3+2)/9 = 0.2235$	0.04474
3	0.2235	$(0.2235^3+2)/9 = 0.2235$	0.0000

Solution of Nonlinear Equations

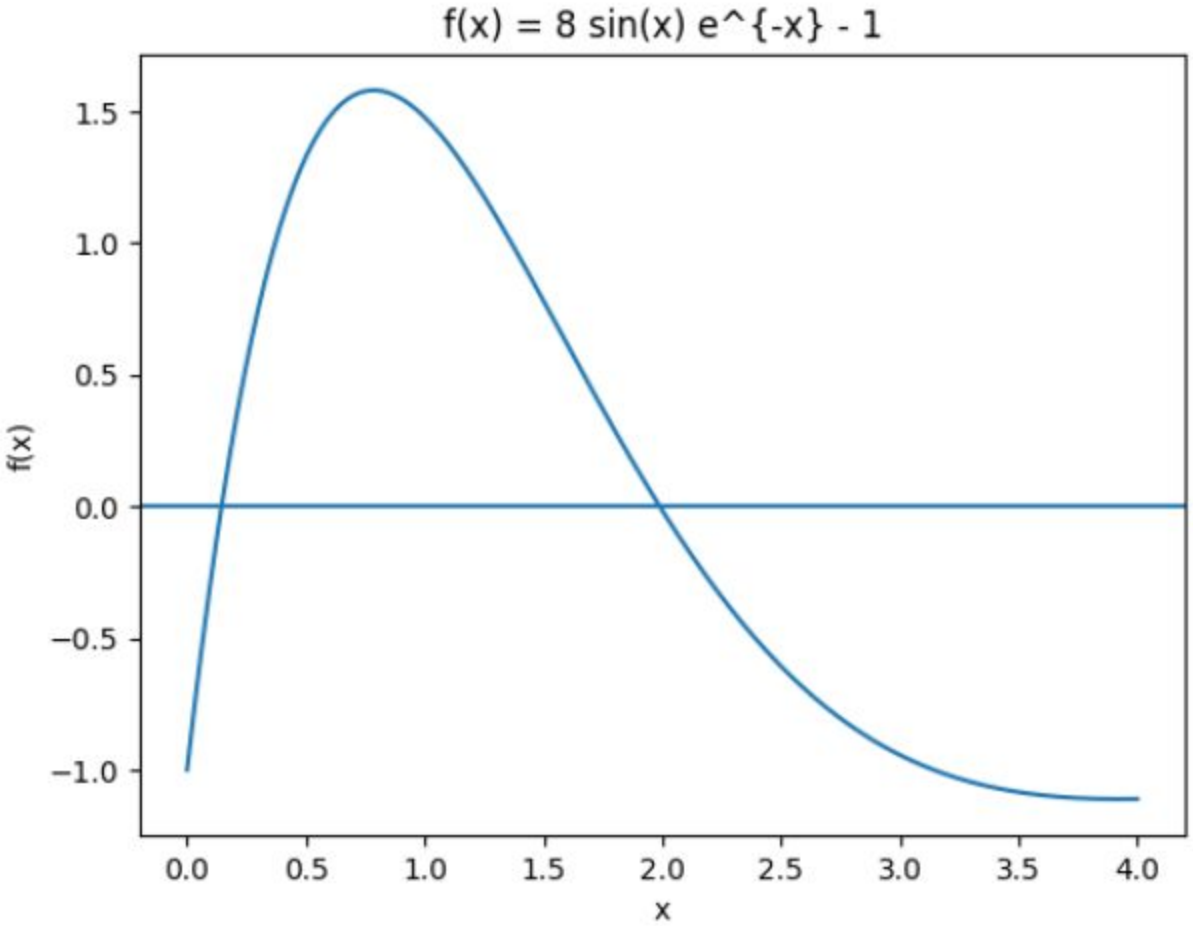
6.10 Determine the lowest positive root of $f(x) = 8\sin(x)e^{-x} - 1$:

(a) Graphically.

(b) Using the Newton-Raphson method (three iterations, $x_i = 0.3$).

(c) Using the secant method (five iterations, $x_{i-1} = 0.5$ and $x_i = 0.4$).

(d) Using the modified secant method (three iterations, $x_i = 0.3$, $\delta = 0.01$).


f(x) = 8 sin(x) e^{-x} - 1
1.5
1.0
0.5
0.0
−0.5
−1.0
f(x)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x

(b) Newton–Raphson Method (3 iterations, $x_0 = 0.3$)

Step 1: Derivative

$$f(x) = 8\sin(x)e^{-x} - 1$$

Using product rule:

$$f'(x) = 8e^{-x}(\cos x - \sin x)$$

Formula

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

i	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	ε_a
1	0.300000	0.751414	3.910431	0.107844	178.2%
2	0.107844	−0.226952	6.367370	0.143487	24.84%
3	0.143487	−0.008952	5.868388	0.145012	1.05%

Root ≈ 0.14501

(c) Secant Method (5 iterations, $x_0 = 0.4,\ x_1 = 0.5$)

Formula:

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

(c) Secant ($x_{i-1} = 0.5, x_i = 0.4$, five iterations)

$$x_{i+1} = x_i - f(x_i)(x_{i-1} - x_i) / [f(x_{i-1}) - f(x_i)]$$

i	x_{i-1}	x_i	$f(x_{i-1})$	$f(x_i)$	x_{i+1}	ε_a
1	0.500000	0.400000	1.326290	1.088279	−0.057239	798.8%
2	0.400000	−0.057239	1.088279	−1.484624	0.206598	127.7%
3	−0.057239	0.206598	−1.484624	0.334745	0.158055	30.71%
4	0.206598	0.158055	0.334745	0.075093	0.144016	9.75%
5	0.158055	0.144016	0.075093	−0.005848	0.145030	0.70%

Root ≈ 0.14503

(d) Modified Secant Method (3 iterations, $x_0 = 0.3,\ \delta = 0.01$)

Formula:

$$x_{i+1} = x_i - \frac{f(x_i)\delta x_i}{f(x_i + \delta x_i) - f(x_i)}$$

(d) Modified secant ($x_0 = 0.3$, $\delta = 0.01$, three iterations)

$$x_{i+1} = x_i - \delta x_i \cdot f(x_i) / [f(x_i + \delta x_i) - f(x_i)]$$

i	x_i	$f(x_i)$	$f(x_i + \delta x_i)$	x_{i+1}	ε_a
1	0.300000	0.751414	0.763094	0.107007	180.4%
2	0.107007	−0.232287	−0.225469	0.143463	25.41%
3	0.143463	−0.009093	−0.000687	0.145015	1.07%

Root ≈ 0.14501

Summary

- The true root is 0.145015. Newton-Raphson and the modified secant method agree closely, because the small perturbation $\delta = 0.01$ makes the modified secant method behave almost like Newton-Raphson, without needing $f'(x)$.
- The secant method needed more iterations since it starts from two points and has no derivative information.